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PII: S1367-9120(25)00198-1
DOI: <https://doi.org/10.1016/j.jseaes.2025.106683>
Reference: JAES 106683

To appear in: *Journal of Asian Earth Sciences*

Received Date: 11 January 2025
Revised Date: 21 May 2025
Accepted Date: 29 May 2025



Please cite this article as: Weng, X-X., Meng, Y-M., Huang, X-W., Shang, L-B., Hu, R-Z., Bi, X-W., Ga- and Ge-bearing minerals and origin of trace element-rich magnetite in the Makeng skarn Fe polymetallic deposit, Eastern China, *Journal of Asian Earth Sciences* (2025), doi: <https://doi.org/10.1016/j.jseaes.2025.106683>

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Ga- and Ge-bearing minerals and origin of trace element-rich magnetite in the Makeng skarn Fe polymetallic deposit, Eastern China

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Abstract

The Makeng skarn Fe polymetallic deposit is the largest Fe deposit in Eastern China. It hosts substantial Fe and Mo resources alongside potential Ga and Ge by-products. However, the distribution and enrichment mechanism of Ga and Ge in this deposit remain poorly understood. Our study reveals that Ga is primarily hosted in pyroxene, amphibole, and magnetite with median contents of 180 ppm, 24 ppm, and 22 ppm, respectively. Germanium is preferentially concentrated in andradite and amphibole with median contents of 50 ppm and 37 ppm, respectively. Sulfides contain negligible Ga and Ge. Four generations of magnetite (Mag-I to Mag-IV) have been identified based on mineral assemblages. Mag-I is mainly associated with andradite and pyroxene, whereas Mag-II is associated with amphibole and chlorite. Mag-III is replaced or filled by pyrite, whereas Mag-IV is associated with pyrrhotite, pyrite and fluorite. Mag-I to Mag-III have median Ga and Ge contents of 13-20 ppm and 4-15 ppm, respectively. In contrast, Mag-IV shows higher median Ga and Ge contents of 113 ppm and 20 ppm, respectively. Gallium incorporates into silicate minerals by substituting for Fe^{2+} / Fe^{3+} or Mn^{2+} , whereas Ge tends to replace Si^{4+} or Al^{3+} . In magnetite, Ga enters via couple substitution, $\text{Ga}^{3+} + \text{Al}^{3+} \leftrightarrow 2\text{Fe}^{3+}$, whereas Ge incorporates through either isomorphic substitution ($\text{Ge}^{2+} \leftrightarrow \text{Fe}^{2+}$) or coupled substitution ($\text{Ge}^{2+} + \text{Mg}^{2+} \leftrightarrow 2\text{Fe}^{2+}$). The enrichment of Ga and Ge in silicate minerals is attributed to their crystal structures and high partition coefficients. The elevated Ga and Ge contents in Mag-IV may result from the superposition of Ga- and Ge-rich fluids, along with lower temperature and higher oxygen fugacity.

Keywords:

Gallium; Germanium; Distribution and occurrence; Makeng Fe deposit; Skarn

1. Introduction

The concepts of critical metals or critical minerals have gained significant attention in recent years (Schulz et al., 2017; Gislev et al., 2018). Gallium (Ga) and germanium (Ge) are recognized as critical metals due to their vital roles in defense, military industries, and high-tech applications, often referred to as the “backbone of industrial electronics” (Erdmann and Graedel, 2011; Frenzel et al., 2016). With increasing global demand, the exploration of new and sustainable resources for these metals has become imperative.

The average crustal abundances of Ga and Ge are approximately 15 ppm and 1.6 ppm, respectively (Taylor and McLennan, 1985), making it difficult for them to form independent deposits. Instead, they typically occur as by-products in association with other mineral deposits (Tu et al., 2004). The United States, China, Russia, and Germany account for about 45%, 41%, 10%, and 2% of global Ge supplies, respectively (Asian metal network). Global Ga resources are mainly distributed in Africa (32%), Oceania (23%), South America (21%), Asia (18%), and other regions (6%). China stands out as a leading producer of Ga, with major deposits concentrated in Inner Mongolia, Sichuan, Guangxi, Guizhou, Henan, and Shanxi provinces (Beazley, 1998; Dun et al., 2019).

Previous studies have documented Ga enrichment in weathering-sedimentary bauxite, coal, and carbonate-hosted Pb-Zn deposits (Dai et al., 2006; Dai et al., 2007; Su et al., 2021; Mi et al., 2022; Qi et al., 2023). In low-temperature environments, Ga and Ge exhibit substantial enrichment in sphalerite, with concentrations reaching up to 600 ppm (Ga) and 1300 ppm (Ge), as observed in MVT-type Pb-Zn deposits in Tennessee of USA, Fankou and Yunnan-Guizhou-Sichuan regions of China (Bonnet, 2014; Meng et al., 2024; Wu et al., 2024). Gallium has also been identified in alunite deposits within high-sulfur epithermal subduction zones (Sahlström et al., 2017), magmatic Fe-Ti-V (Song et al., 2024), volcanogenic massive sulfide (Safina et al., 2018), rare metal pegmatite, and alkaline pegmatite deposits (Černý, 1991; Wen et al., 2020; Yang et al., 2024).

Gallium and Ge exhibit a strong chemical affinity for Fe, with ionic radius of Ga^{3+} closely resembling that of Fe^{3+} in a six-coordinate environment, facilitating its incorporation into Fe-bearing minerals. Meanwhile, Ge favors an eight-coordinate environment and can substitute for Fe^{3+} in the Fe oxides, leading to its enrichment in iron oxide minerals, particularly within banded iron formations (BIF) (Bernstein, 1985; Tu et al., 2004; Höll et al., 2007). For example, the Lahn-Dill BIF deposit in Germany has average Ge concentrations of 40 ppm (up to 100 ppm) in magnetite and 8 ppm (up to 20 ppm) in hematite (Schrön, 1968; Höll et al., 2007). Gallium is primarily associated with iron oxide apatite (IOA) deposits, such as those in Ningwu and Luzong districts within the Middle to Lower Reaches of the Yangtze River Metallogenic Belt (Zhou et al., 2020) and iron oxide-copper-gold (IOCG) deposits in the Kangdian Metallogenic

Belt of Southwest China (Zhao and Zhou, 2011; Dare et al., 2014; Chen et al., 2015). Additionally, Ga and Ge enrichment has been documented in skarn Fe deposits. For instance, the Chengchao skarn Fe deposit in eastern Hubei, China, hosts magnetite with Ga contents ranging from 2.5 to 42 ppm, representing an estimated resource of approximately 1482 metric tons (Hu et al., 2014; Zhou et al., 2020). The Fenghuangshan skarn Cu-Fe-Au deposit in Tongling, China shows up to 97 ppm Ga and 29 ppm Ge in magnetite (Huang et al., 2016). However, a systematic investigation into the distribution and occurrence of Ga and Ge in skarn deposits remains lacking.

In this study, we systematically investigated the distribution of Ga and Ge in different types of ores and minerals such as silicates, iron oxides, and sulfides from the Makeng skarn Fe polymetallic deposit in Fujian, China. The substitution mechanisms of Ga and Ge in different minerals were elucidated by element correlation analysis. By comparing published Ga and Ge data of magnetite from global skarn deposits, we explored the key factors controlling their enrichment in magnetite. This research advances understanding of critical metal behavior in hydrothermal systems and offers practical guidance for comprehensive utilization of polymetallic resources.

2. Geological background

The South China Plate exhibits a distinct west-to-east structural zonation, consisting of three major tectonic units, the Yangtze Block, the Jiangnan Orogenic Belt, and the Cathaysia Block (Fig. 1a), which is also referred to as the South China Fold Belt (Shu et al., 2021). Within this framework, Fujian Province occupies the southeastern portion of the South China Fold Belt and can be subdivided into three principal tectonic domains, the northwestern Fujian uplift, the Yongmei depression in southwestern Fujian, and the volcanic fault depression belt in eastern Fujian (Wei et al., 1997). Of these, the Yongmei depression zone represents the most significant Fe polymetallic metallogenic belt in South China (Fig. 1b), accounting for nearly 98% of discovered Fe ore reserves in Fujian (Zuo et al., 2012; Zuo et al., 2015). This metallogenic belt hosts several important deposits including Makeng, Luoyang, Dapai, Yangshan, Pantian, and Zhongjia, with the Makeng deposit being the largest in terms of both size and economic importance (Wu and Wu, 1997; Zhang, 2010).

The tectonic architecture of southwestern Fujian is characterized by complex deformation patterns, primarily controlled by three major fault systems, Ninghua-Nanping, Shanghang-Yunxiao, and Zhenghe-Dapu structural belts (Fig. 1b). This region displays extensive development of thrust nappe structures and fold systems. The Makeng Fe deposit, located along the western margin of the Zhenghe-Dapu fault belt, occupies a particularly favorable structural position for Fe polymetallic mineralization (Zhang, 2000; Mao et al., 2001; Zhang and Zuo, 2015). Stratigraphically, southwestern Fujian preserves a nearly complete geological record with the notable absence of

Silurian, Middle-Lower Devonian, and Paleogene strata (Fig. 1b) (Li and Liang, 1987; Zhang, 2000). The strata in this region can be classified into three major discontinuous chronological rock series based on stratigraphic characteristics, magmatic intrusion, and metamorphic deformation. The Pre-Devonian basement series, composed of late Archean to early Proterozoic continental sedimentary volcanic flysch formations, overlain by Sinian and early Paleozoic (Z-O) marine sedimentary rocks. The Late Paleozoic-Middle Triassic epicontinental sea sedimentary cover, predominantly developed in the Yong'an-Meixian fold belt. The Meso-Cenozoic continental clastic-volcanic rock series comprises a mixture of continental clastic materials, volcanic eruptive products, and sedimentary deposits (Zhang, 2000; Mao et al., 2001; Zhang et al., 2012b). The region features numerous Yanshanian, Indosinian, Hercynian, and Caledonian granitoids with surface exposures dominated by Yanshanian and Hercynian mafic rocks, particularly diabase. Recent geochronological studies using SHRIMP and laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) zircon U-Pb dating of granitoid associated with Fe deposits at Zhongjia, Dapai, Luoyang, and Pantian have yielded ages of 132-146 Ma (Yang et al., 2008; Zhang et al., 2018), confirming the importance of Yanshanian magmatism in the regional metallogeny.

3. Deposit geology

The dominant rock units within Makeng deposit exhibit a northeast-southwest orientation (Fig. 2a), adhering to the principles of stratigraphic succession (Zhang and Zhang, 2014; Zhang et al., 2015a). The basal unit is identified as the Lower Carboniferous Lindi Formation, which predominantly consists of continental clastic rocks, which is overlain by the Upper Carboniferous Huanglong Formation composed of a series of pyroclastic rocks. Subsequent units include carbonate and offshore shallow sea sediments spanning from the Upper Carboniferous Chuanshan Formation to the Middle Permian Qixia Formation, followed by sandstone and shale from the Middle Permian Wenbisha and Tongziyan Formations. Additionally, minor exposures of Ordovician-Silurian strata are present in the northeastern part of the study area (Fig. 2).

The Yanshanian Dayang granite and Juzhou granite are distributed on either side of Fe ore body. Despite a surface separation of merely 1 km², these granitic bodies extend to considerable depths and are intricately interconnected within the Lindi Formation (Zhang, 2010; Zhang et al., 2012b; Zhang and Zhang, 2014). The Juzhou-Dayang granitic intrusion exhibits different phase zonation after differentiation. The southern portion consists of flesh-red medium fine-grained microcline feldspar granite facies, whereas the northern portion is characterized by grayish white to light flesh-red, medium-grained biotite syenogranite. Both facies share similar mineral compositions, dominated by potassium feldspar, followed by quartz, plagioclase, and minor biotite,

with accessory minerals including magnetite, sphene, and apatite (Lin, 2011; Zhang et al., 2012b). The main ore body of the Makeng Fe deposit occurs in the external contact zone of Juzhou-Dayang granite (Fig. 2b), specifically within the tectonic fracture zone separating the Huanglong, Jingshe and Lindi Formations (Fig. 2b) (Zhang et al., 2012b; Zhang and Zhang, 2014). U-Pb dating of andradite and Re-Os dating of molybdenite constrain the Fe mineralization to 133-130 Ma, consistent with the timing of granite emplacement (Zhang et al., 2012a, b; Wang et al., 2023).

The mining area exhibits extensive wall rock alteration, manifested by the pervasive development of calcic skarn, which formed through metasomatic reactions between granitic or diabase magmas and the carbonate-rich strata (Fig. 2) (Zhao et al., 1983). The Makeng skarn Fe deposit follows a mineralization pattern typical of skarn Fe deposits, with four distinct stages defined by mineral assemblages and fluid evolution (Zhao et al., 1983; Zhao and Lin, 1990; Zhang, 2012; Li et al., 2016). Early prograde skarn stage is characterized by a mineral assemblage dominated by diopside euhedral andradite and minor magnetite. Fluid inclusions in garnet and diopside have homogenization temperatures of 440-600 °C, with salinities of 6-24 wt% and 32-44 wt% NaCl_{equiv}, respectively (Zhang et al., 2013; Yang et al., 2017). The retrograde stage involves the replacement of early-forming anhydrous silicate minerals into amphibole series minerals, accompanied by increased magnetite abundance. Two types of fluid inclusions in diopside exhibit homogenization temperatures of 448-564 °C and 501-594 °C, respectively, with the salinities of 27-48 wt% and 2-7 wt% NaCl_{equiv}, respectively (Zhang et al., 2013; Yang et al., 2017). The quartz-metal sulfide stage is dominated by metal sulfides such as molybdenite, pyrite and pyrrhotite. Fluid inclusions in quartz show the homogenization temperatures and salinities of 160-400 °C and 0-4 wt% NaCl_{equiv}, respectively (Zhang et al., 2013). The carbonatite stage is characterized by calcite, fluorite and quartz vein cutting the earlier magnetite and sulfides. Fluid inclusions in calcite and fluorite indicate the homogenization temperatures of 182-343 °C and salinities of 2-20 wt% NaCl_{equiv} (Yang et al., 2017).

The ore body generally aligns with the roof and floor strata or the surrounding rock, exhibiting a layered structure (Fig. 2b). It extends approximately 3000 m in length, 100-1200 m in width, and 10-100 m in thickness (Zhang, 2010, 2012). The Makeng deposit comprises eastern and western ore sections, with proven Fe ore reserves of 434 Mt averaging 38%, 0.08 Mt Mo grading at 0.04-0.24 %, and Ga and Ge grades of 0.0023-0.0027 % and 0.0017 %, respectively (Jiang, 2009; Zhang, 2010; Jiang et al., 2020).

4. Sampling and analytical methods

4.1. Sampling

Samples for this study were collected from the middle sections of exploration lines 70 and 72 at the Makeng Fe deposit, with a focus on the oxidation zone, the orebody, and the mineralized surrounding rock. A total of 36 samples were obtained, of which 18 representative samples were selected for preparing powder and thin sections.

4.2. Solution ICP-OES and ICP-MS analyses

Major and trace element contents were determined at State Key Laboratory of Ore Deposit Geochemistry (SKLOGD), Institute of Geochemistry, Chinese Academy of Sciences (IGCAS), Guiyang, utilizing Agilent 720 ICP-OES and Jena PlasmaQuant MS Elite ICP-MS, respectively. The specific instrument parameters were implemented as described in Qi and Gregoire (2000) and Huang et al. (2023). About 50 mg of whole-rock powder from each sample was digested in a PTFE-coated high-pressure steel bomb using a mixed acid solution of concentrated HF and HNO₃ at 200 °C for 12 h. After evaporation to dryness, Rh solution was added in insoluble residues, which were re-dissolved in concentrated HNO₃ at 140 °C for 3 h. The final solution was split and diluted into two aliquots for ICP-MS and ICP-OES analyses, respectively. Quantitative contents were calibrated using external solution standards, with Rh as an internal standard to correct for instrumental drift and matrix effects (Qi and Gregoire, 2000).

4.3. X-ray diffractometer

Mineral composition analysis was conducted using an Empyrean X-ray diffractometer (Malvern Panalytical) at the SKLOGD, IGCAS. Homogenized powder samples were front-loaded into aluminum sample holders and secured in the dedicated sample stage slot. The instrument features a ceramic-metal tube with a maximum power output of 2.2 kW and a goniometer with a minimum step accuracy of 0.0001°, using Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$) generated at 40 kV and 40 mA (Lu et al., 2024). Scans were performed over a 5-70° 2 θ range in continuous scan mode with 0.02° step size and 1 s/step counting time. The collected diffraction patterns were analyzed using JADE software with phase identification conducted through the ICDD PDF-4+ database and semi-quantitative analysis performed using Rietveld refinement.

4.4. SEM

Petrographic observations were conducted using a JEOL JSM-7800F field emission scanning electron microscope (FE-SEM) at the SKLOGD, IGCAS. The instrument was operated at an acceleration voltage of 10-15 kV with a beam current of 10 nA, employing a 1 μm beam spot diameter for high-resolution imaging. Energy-

dispersive X-ray spectroscopy (EDS) analysis was performed with a 20-second acquisition time per spectrum to ensure reliable elemental characterization. The SEM was calibrated using copper standard (Cu K α = 8.04 keV) prior to analysis, and sample surfaces were carbon-coated to enhance conductivity.

4.5. LA-ICP-MS trace element analysis

The trace elements compositions of silicates, sulfide minerals and magnetite were analyzed by LA-ICP-MS at the SKLOG, IGCAS using three complementary analytical systems.

Silicates were analyzed using a GeoLasPro 193 nm ArF excimer laser combined with an Agilent 7900 ICP-MS. The system operated at a laser frequency of 6 Hz with a beam spot diameter of 40 μ m, employing 96-second single measurement durations. Data reduction was performed using ICPMSDataCal software (Liu et al., 2008), with NIST SRM 610 and NIST SRM 612 as primary calibration standards and BCR-2G as a quality control reference material.

Magnetite was analyzed by a New Wave Research (NWR) 213 nm laser ablation system connected to an Agilent 7700x ICP-MS, utilizing a 30-second background measurement followed by 50-second ablation under helium carrier gas (0.8 L/min) mixed with argon (0.9 L/min) through a T-joint interface. Calibration standards include USGS reference glasses (GSE-1G, BCR-2G, and BHVO-2G) with BC-28 and GOR128-G as monitoring standards (Dare et al., 2012; Huang et al., 2015; Huang et al., 2024), and data were processed offline using the ICPMSDataCal software (Liu et al., 2008).

Sulfide analyses were conducted using the ASI RESOLution-LR-S155 excimer laser system coupled with the Agilent 7700x ICP-MS. The system configuration included argon (0.9 L/min) and helium (0.35 L/min) mixed in the ablation cell, with the aerosol transported to the ICP. The laser working parameters were set to a frequency of 6 Hz and a beam spot of 26 μ m, employing acquisition cycles of 30-second background measurements followed by 60-second sample ablation. External calibration standards include STDGL3, GSD-1G, and natural pure pyrite (Py), whereas the sulfide international standard sample MASS-1 served as a quality control. Trace element quantification employed a multi-external standard and total normalization method (Liu et al., 2008), where STDGL3 corrected for chalcophile elements (e.g., S, Cu, Ag, Au, Zn, Cd, Ga, Ge, In, Sn, Tl, Pb) and GSD-1G corrected for lithophile elements (e.g., Si, Sc, Ti, Mn, Rb, Sr, Y, Nb, Mo, Ta, W) (Danyushevsky et al., 2011).

5. Results

5.1. Petrography

Based on XRD mineral proportions (Appendix Table S1) and ore structure, five types of ores are defined (Type 1-5). Type-1 ores consist of 38-72% andradite, 14-60% magnetite, and minor amphibole, quartz, and fluorite. The anhedral magnetite aggregates are filled around the relatively euhedral andradite to form a massive structure (Fig. 3d). Type-2 ores are predominantly composed of magnetite (13-93%), with subordinate hydrous silicate minerals (e.g., amphibole) and minor andradite. Magnetite is fine-grained and commonly crosscut by irregular quartz-calcite-fluorite veins, while hematite occurs as fine-grained aggregates (Fig. 3e). Amphibole (17-30%) is the dominant mineral in Type-3 ores, accompanied by minor magnetite (12-38%), chlorite, and kaolinite (Fig. 3f). Chlorite, calcite, and quartz occur as veinlets in magnetite to form banded structure (Fig. 3f). Type-4 ores are composed of quartz (27–34%), magnetite (22–36%), calcite (10–23%), and sulfides (4–19%) (Fig. 3g, h). Pyrite, chalcopyrite, and pyrrhotite are arranged in parallel bands within the ore, and minor hematite filling gaps in the ore (Fig. 3g, h). Type-5 ores are predominantly composed of calcite + quartz (51–87%), chlorite (10–14%), magnetite (1–8%), with minor fluorite and molybdenite (Fig. 3i). Magnetite occurs as disseminated grains or veinlets associated with chlorite within the ore, with molybdenite veins crosscutting the ore (Fig. 3i).

The non-metallic minerals mainly include andradite, pyroxene, amphibole, chlorite, fluorite, calcite, and quartz. Andradite, characterized by its distinct size, frequently occurs in association with magnetite, pyroxene and amphibole which are cut by quartz veins or replaced by chlorite along the margin (Fig. 4a). In certain cases, late-stage minerals are metasomatized along the zoning of andradite (Fig. 4b). Anhedral pyroxene and amphibole primarily occur as incomplete replacements of earlier stage andradite, being replaced by later stage minerals such as fluorite, quartz, and calcite. Extensive alteration leads to the formation of chlorite (Fig. 4e), which is often distributed in veins, cracks, or within metasomatic magnetite.

Metallic minerals are dominated by magnetite, with minor amounts of molybdenite, pyrite, pyrrhotite, chalcopyrite, hematite, and sphalerite (Fig. 4). Sphalerite is associated with pyrite, chalcopyrite, and mainly occurs in quartz veins (Fig. 4a, g), also showing exsolution and metasomatic textures related to pyrite and chalcopyrite. Molybdenite predominantly occurs in particles measuring 20-60 μm , both minerals are primarily found within quartz veins (Fig. 3i). Hematite occurring as vane-shaped or needle-shaped crystals, fills the interstitial spaces between silicate minerals and magnetite (Fig. 3e).

Based on paragenetic mineral associations, four distinct generations of magnetite (Mag-I to Mag-IV) have been identified in the Makeng deposit. The first-generation

magnetite (Mag-I) is subhedral and is intimately associated with andradite and pyroxene, forming elongated textures (Fig. 4a-c). The second-generation magnetite (Mag-II) has oscillatory zoning or is subhedral associated with andradite, amphibole and chlorite (Fig. 4d-f). The third-generation magnetite (Mag-III) is commonly replaced by amounts of sulfides such as pyrite (Fig. 4g-i), whereas the fourth-generation magnetite (Mag-IV) is associated with pyrrhotite, pyrite and fluorite (Fig. 4j-l).

5.2. Major and trace elements of ores

The complete chemical composition data for ores and altered rocks are listed in Appendix Table S2. Gallium concentrations remain relatively uniform across all ore types (Type 1-5), with median values ranging from 14 to 24 ppm. Germanium exhibits a median content of 12 to 29 ppm in Type 1-4 ores, but shows an obvious decrease to 7 ppm in Type 5 ores that correlates with their lower magnetite content. We divide the whole rock samples into two groups based on ore grade of Fe. Group 1 is Fe-rich ores with TFe content higher than 45 wt.%, whereas group 2 is Fe-poor ores and altered rocks (TFe < 45 wt.%). Group 1 exhibits TFe contents ranging from 45 to 59 wt.%, with a median of 54 wt.%. Whole-rock Ga concentrations in Group 1 range from 14 to 26 ppm (median 21 ppm), and Ge concentrations ranged from 24 to 31 ppm (median 28 ppm). Group 2 contains 2-36 wt.% TFe, with the Ga content of 7-30 ppm (median: 17 ppm) and Ge content of 1-19 ppm (median: 16 ppm).

Statistical analysis reveals Ge exhibits strong positive correlations with Fe, V, Cr and Ni, along with moderate negative correlations with Al and Ti (Fig. 5). Additionally, Ge shows a distinct negative correlation with Mn content (Fig. 5).

5.3. Trace element composition of minerals

The complete LA-ICP-MS results are provided in Appendix Table S3. Andradite contains Ga contents of 2-24 ppm (median: 12 ppm) and Ge contents of 4-180 ppm (median: 50 ppm). Pyroxene shows higher Ga (53-196 ppm, median: 180 ppm) but lower Ge contents (5-76 ppm, median: 5 ppm), while amphibole exhibits intermediate Ga (1-104 ppm, median: 24 ppm) and Ge (7-58 ppm, median: 37 ppm) contents (Fig. 6).

Mag-I, Mag-II, and Mag-III show comparable Ga contents (10-38 ppm, median: 13-20 ppm), obviously lower than Ga in Mag-IV (99-146 ppm, median: 113 ppm) (Fig. 6b). Four generations of magnetite have similar median Ge contents (5-12 ppm), but Ge in Mag-IV exhibited a broader Ge range (5-40 ppm) (Fig. 6b).

Molybdenite, sphalerite, and pyrite have Ga contents ranging from 0.07 to 3.36

ppm, while pyrrhotite and chalcopyrite have Ga contents below the detection limits (Fig. 6a). Sphalerite, pyrite, and pyrrhotite have Ge contents of 0.74-3.76 ppm, while molybdenite and chalcopyrite have Ge contents below the detection limits (Fig. 6a).

6. Discussion

6.1. Distribution of Ga and Ge

The Makeng deposit exhibits significant enrichment of Ga and Ge, with whole rock concentrations reaching up to 30 ppm and 31 ppm, respectively, substantially exceeding their crustal abundances of 15 ppm and 1.6 ppm, respectively (Taylor and McLennan, 1985). Gallium maintains relatively stable concentrations (14-24 ppm median) across all ore types, suggesting its dispersion in major silicates and oxides. The positive correlation of Ga with V, Cr, and Ni (Fig. 5) indicates that Ga is mainly hosted in mafic silicate minerals or Fe oxides. Germanium is lower in Fe-poor ores and the positive correlation between Ge and Fe (Fig. 5) indicate that Ge is mainly hosted by Fe-bearing minerals.

LA-ICP-MS results reveal that Ga and Ge are mainly distributed in andradite, pyroxene, amphibole, and magnetite (Fig. 6). The median contents of Ga and Ge are 12 ppm and 50 ppm in andradite, 180 ppm and 5 ppm in pyroxene, and 24 ppm and 37 ppm in amphibole, respectively. The distribution of Ga in andradite and pyroxene in the Makeng deposit differs from trends observed in other deposits. In most skarn deposits, Ga concentrations are higher in garnet than in pyroxene. For example, in the Nanyangtian W-Sn-Fe skarn deposit of southeastern Yunnan, the median Ga concentration in pyroxene and garnet is ~2 ppm and 26 ppm, respectively, with no detectable Ge (Niu, 2021). Similarly, in the Zhuchong skarn Fe deposit of the Middle-Lower Yangtze River Metallogenic Belt in eastern China, the median Ga contents in garnet, diopside, and actinolite are 32 ppm, 8 ppm, and 11 ppm, respectively, whereas Ge concentrations are 20 ppm, 3 ppm, and 5 ppm, respectively (Fig. 10) (Liang et al., 2023b). In contrast, the distribution of Ge in the Makeng deposit aligns with observations from other deposits, where garnet consistently exhibits higher Ge concentrations than pyroxene. It is worth noting that amphibole in the Makeng deposit also contains measurable amounts of both Ga and Ge, with median concentrations falling between those observed in garnet and pyroxene.

Germanium content in Mag-I to Mag-III is relatively stable at a median of 9 ppm but increases to 20 ppm in Mag-IV (Fig. 6). Gallium is evenly distributed in Mag-I to Mag-III, with a median of 22 ppm, but rises significantly to 113 ppm in Mag-IV (Fig. 6). This evolutionary trend is different from other skarn deposits. In the Fenghuangshan skarn Cu-Fe-Au deposit, magnetite is classified into three generations corresponding to early retrograde skarn, late retrograde skarn, and carbonate stages (Huang et al., 2016).

The first and second generations of magnetite exhibit comparable median Ga contents (26 ppm and 24 ppm, respectively), which then decrease to 9 ppm in the third generation, while Ge content remains relatively stable at ~5 ppm. Similarly, in the Zhuchong skarn Fe deposit, the median Ga content in magnetite declines from 80 ppm in the skarn stage to 55 ppm in the oxide stage and 59 ppm in the sulfide stage, while Ge content remains unchanged at ~4–5 ppm (Liang et al., 2023b). These findings suggest that Ga in magnetite generally decreases during hydrothermal evolution, whereas Ge remains relatively stable. Therefore, the abrupt increase of Ga and Ge in Mag-IV from the Makeng deposit indicate the uniqueness for the formation of this generation of magnetite.

6.2. Substitution mechanisms of Ga and Ge in minerals

Recent studies have investigated the distribution of trace elements such as Ga and Ge in various minerals using LA-ICP-MS, and demonstrated that these elements primarily exist in silicate minerals, oxides, and sulfides through the substitution of other elements (Cook et al., 2009; Ye et al., 2011; Luo et al., 2021; Meng et al., 2024). The highest Ga and Ge contents in silicate minerals reached 196 ppm and 180 ppm, respectively, whereas the highest Ga and Ge contents in magnetite were 146 ppm and 40 ppm, respectively. The concentrations of Ga and Ge are significantly lower than those observed in any known independent Ga or Ge minerals (Zhang et al., 2003; Foley et al., 2017; Sahlström et al., 2017; Meng et al., 2023). Time-resolved LA-ICP-MS signal spectra lack obvious peaks of Ga and Ge in andradite, pyroxene, amphibole, and magnetite (Fig. 7). Furthermore, no independent Ga or Ge minerals were identified during SEM observation. These results demonstrate that both elements occur dominantly through isomorphic substitution in the crystal structures of host silicates and oxides.

Andradite is a typical silicate mineral with an island structure, represented by the chemical formula of $A_3B_2(SiO_4)_3$, where A denotes bivalent cations (e.g., Mg^{2+} , Fe^{2+} , Mn^{2+} and Ca^{2+}) and B represents trivalent cation (e.g., Al^{3+} , Fe^{3+} , Cr^{3+} , Ti^{3+}). The negative correlation between Fe and Ga + Mn (Fig. 8a) suggests that Ga may substitute for Fe in andradite through the coupled substitution mechanism, $3(Ga^{3+} + Mn^{2+}) \leftrightarrow 5Fe^{3+} + \square$. Similarly, the negative correlation between Ge and Si (Fig. 8b) indicates that Ge primarily replaces Si within the andradite lattice via the isomorphic substitution, $Ge^{4+} \leftrightarrow Si^{4+}$.

Pyroxene is a single-chain silicate mineral with general formula of $AB[Si_2O_6]$ ($A=Na^+$, Ca^{2+} , Mn^{2+} , Fe^{2+} , Mg^{2+} , Li^+ ; $B=Mn^{2+}$, Fe^{2+} , Fe^{3+} , Al^{3+} , Cr^{3+} , Ti^{4+}). The negative correlation between Ga and Fe (Fig. 8c) in pyroxene suggests that Ga is incorporated into the pyroxene lattice through the substitution $2Ga^{3+} + \square \leftrightarrow 3Fe^{2+}$. A strong positive correlation between Ge and Fe, and negative correlation between Ge +

Fe and Al in pyroxene (Fig. 8d-e), indicates that Ge exhibits a strong affinity for Fe (Bernstein, 1985; Höll et al., 2007; Frenzel et al., 2014) and enters the pyroxene lattice via the coupled substitution $\text{Ge}^{4+} + \text{Fe}^{2+} \leftrightarrow 2\text{Al}^{3+}$.

Amphibole, a double-chain silicate mineral, has the ideal chemical formula of $\text{A}_0\sim_1\text{B}_2\text{C}_5\text{T}_8\text{O}_{22}(\text{OH}, \text{F}, \text{Cl})_2$ (A=Na, K; B=Na, Zn, Li, Ca, Mn, Fe^{2+} , Mg; C=Mg, Fe^{2+} , Mn, Al, Fe^{3+} , Cr, Ti; T=Si, Al, Ti). Gallium exhibits a strong affinity for Fe^{3+} (Tu et al., 2004). The ionic radius of Mn^{2+} ranges from 0.67 to 0.83 Å, depending on coordination environment, closely resembling that of Ga^{3+} (0.62 Å). The positive correlation between Ga and Fe (Fig. 8f) and negative correlation between Ga + Fe and Mn (Fig. 8g) suggest that Ga incorporates into the amphibole lattice in combination with Fe through $\text{Ga}^{3+} + \text{Fe}^{3+} + \square \leftrightarrow 3\text{Mn}^{2+}$ (Liang et al., 2023a). The negative correlation between Ge and Si (Fig. 8h) indicates that Ge primarily replaces Si within the amphibole structure.

Magnetite exhibits a cubic inverse spinel structure (AB_2O_4) with different cationic sites, where the A site is occupied by divalent cations (e.g., Mg^{2+} , Fe^{2+} , Ni^{2+} , Mn^{2+} , Co^{2+} , Zn^{2+}) and B site hosts trivalent cations (e.g., Al^{3+} , Fe^{3+} , Cr^{3+} , V^{3+} , Mn^{3+} , Ga^{3+}) (Fleet, 1981; Meng et al., 2017). The correlation between Ga and Fe is not obvious for all generations of magnetite (Mag-I to Mag-IV) (Fig. 9a). However, a weak positive correlation between Ga and Al, along with negative correlation between Ga + Al and Fe (Fig. 9b, c), suggests that Al and Ga incorporate into the magnetite structure through the coupled substitution mechanism $\text{Ga}^{3+} + \text{Al}^{3+} \leftrightarrow 2\text{Fe}^{3+}$.

In the Makeng skarn Fe-polymetallic deposit, Mag-IV is associated with abundant pyrrhotite and pyrite, indicating formation under reduced, sulfide-rich hydrothermal conditions that favor Ge^{2+} speciation (Bernstein, 1985; Höll et al., 2007; Yang et al., 2017). Germanium incorporation in magnetite occurs through two distinct substitution mechanisms: direct isovalent substitution ($\text{Ge}^{2+} \leftrightarrow \text{Fe}^{2+}$), evidences by the weakly negative correlation between Ge and Fe (Fig. 9d); and coupled substitution ($\text{Ge}^{2+} + \text{Mg}^{2+} \leftrightarrow 2\text{Fe}^{2+}$), supported by the positive correlation between Ge and Mg (Fig. 9e) and the negative correlation between Ge + Mg and Fe (Fig. 9f).

6.3. Enrichment mechanism

6.3.1. Crystal structure of minerals

In the Makeng Fe deposit, Ga is primarily hosted by silicate minerals (amphibole and pyroxene) and magnetite (particularly in Mag-IV), while Ge is mainly concentrated in andradite and amphibole, with secondary occurrences in pyroxene and magnetite.

Germanium exhibits geochemical behavior analogous to silicon in crustal environments, primarily due to their similar ionic radii (Ge^{4+} : 0.53 Å; Si^{4+} : 0.4 Å) and

tetrahedral coordination preferences (Liu et al., 1984; Bernstein, 1985; Tu et al., 2004; Höll et al., 2007). Similar mineral-melt partition coefficients for Ge in pyroxene (1.6–3.8) and amphibole (1.5–2.5) (Ewart and Griffin, 1994) demonstrate that Ge distribution among silicates is not solely controlled by partitioning behavior. Instead, Ge concentrations systematically decrease with increasing silicate structural complexity: from isolated (e.g., topaz, andradite) to framework silicates (e.g., quartz, feldspars), with intermediate values in paired (e.g., epidote), chain (e.g., amphibole), and sheet silicates (e.g., micas) (Bernstein, 1985; Tu et al., 2004; Höll et al., 2007). This structural control explains the observed Ge enrichment sequence in Makeng ($\text{Ge}_{\text{andradite}} > \text{Ge}_{\text{amphibole}} > \text{Ge}_{\text{pyroxene}}$). The mineral-melt partition coefficients of Ga in pyroxene (0.2–0.7) and amphibole (0.4–2.4) (Ewart and Griffin, 1994), align with the observed Ga enrichment in amphibole relative to pyroxene at Makeng. Because Ga^{3+} (0.62 Å) and Ge^{2+} (0.73 Å) have similar ionic radii with Fe^{3+} (0.63 Å) and Fe^{2+} (0.76 Å), respectively, supporting the possibility of Ga and Ge substituting for Fe within the magnetite lattice in the tetrahedral sites or octahedral sites (Höll et al., 2007; Nadoll et al., 2014).

6.3.2 Temperature and fluid composition

The compiled contents of Ga and Ge in magnetite from skarn deposits are illustrated in Fig. 10. Systematic comparison reveals that Ga is notably enriched (average 20–60 ppm) in magnetite from the Zhangmatun, Fenghuangshan, Huangshaping, Beiya, Baixiangshan, and Zhongjiu deposits (Huang et al., 2016; Sun et al., 2017; Ding et al., 2018; Sun et al., 2019) (Fig. 10). Magnetite from the Beiya deposit contains elevated Ge (avg. 59 ppm) (Sun et al., 2017), while magnetite from other deposits typically contains <15 ppm Ge (Fig. 10).

The Mag-IV from the Makeng deposit exhibits significantly higher concentrations of Ga and Ge compared to other deposits (Fig. 10). Studies on fluid inclusions of the Makeng deposit reveal that the homogenization temperatures of the early prograde skarn, retrograde skarn, quartz-sulfide, and carbonate stages are 440–600 °C, 448–564 °C and 501–594 °C, 160–400 °C, and 182–343 °C, respectively (Zhang et al., 2013; Yang et al., 2017), with decreasing temperature during fluid evolution. The obvious correlation between V and Ti and decreased V and Ti contents from Mag-I to Mag-IV, indicating decreasing temperature and increasing oxygen fugacity (Nielsen et al., 1994; Nadoll et al., 2014). From Mag-I to Mag-IV, Ti and V contents decrease and Ga and Ge contents increase in Makeng deposit (Fig. 11), indicating that relatively low temperature and high oxygen fugacity may be beneficial for Ga and Ge enrichment in the Makeng deposit. However, the negative correlation of Ga and Ge with Ti and V is not applicable for magnetite from other skarn Fe deposits (Fig. 11). For example, a decreasing trend of Ga in magnetite is observed from the early to late stages in the Fenghuangshan and Zhuchong deposits (Huang et al., 2016; Liang et al., 2023b). This

indicates that temperature and high oxygen fugacity are not the major factors controlling Ga enrichment in magnetite. Alternatively, other factors such as Ga and Ge in the hydrothermal fluids play an important role. Previous studies have documented multiple magmatic-hydrothermal events in the Makeng deposit (Yang et al., 2017; Zhao et al., 2020; Yi et al., 2021; Wang et al., 2023). It is thus inferred that the enrichment of Ga and Ge in Mag-IV likely results from the superposition of late-stage, Ga-Ge-enriched fluids. Further research is required to explore how the composition and physicochemical properties of ore-forming fluids influence the enrichment of Ga and Ge in skarn deposits.

7. Conclusion

The Makeng Fe deposit in Fujian Province contains economically significant concentrations of critical metals, with ores containing 15-21 ppm Ga (median: 19 ppm) and 5.7-25 ppm Ge (median: 17 ppm). Our mineralogical and geochemical investigations reveal several key findings regarding Ga and Ge occurrence and enrichment mechanisms. Gallium and Ge occur exclusively in solid solution within host minerals, with no discrete mineral phases identified. Significant enrichment occurs in both silicate minerals (andradite, pyroxene, amphibole) and magnetite, particularly Mag-IV in the late stage. Gallium enters andradite, pyroxene and amphibole via isomorphic or coupled substitution mechanisms, $3(\text{Ga}^{3+} + \text{Mn}^{2+}) \leftrightarrow 5\text{Fe}^{3+} + \square$ (andradite), $2\text{Ga}^{3+} + \square \leftrightarrow 3\text{Fe}^{2+}$ (pyroxene), and $\text{Ga}^{3+} + \text{Fe}^{3+} + \square \leftrightarrow 3\text{Mn}^{2+}$ (amphibole). Germanium enrichment follows silicate structural complexity (andradite > amphibole > pyroxene), whereas Ga distribution correlates with mineral-specific partition coefficients. Gallium enters magnetite through coupled substitution, $\text{Ga}^{3+} + \text{Al}^{3+} \leftrightarrow 2\text{Fe}^{3+}$, whereas Ge enters the lattice through both isomorphic substitution ($\text{Ge}^{2+} \leftrightarrow \text{Fe}^{2+}$) and coupled substitution ($\text{Ge}^{2+} + \text{Mg}^{2+} \leftrightarrow 2\text{Fe}^{2+}$). The enrichment of Ga and Ge in magnetite of specific generation is likely related to overlapping of Ga- and Ge-rich fluids, as well as conditions of lower temperature and higher oxygen fugacity.

CrediT authorship contribution statement

Xiang-Xue Weng: Conceptualization, Validation, Data analysis, Visualization, Writing - original draft. **Yu-Miao Meng:** Conceptualization, Funding acquisition, Writing - review & editing. **Xiao-Wen Huang:** Resource, Methodology, Funding acquisition, Writing - review & editing. **Lin-Bo Shang:** Validation, Supervision, Writing - review & editing. **Rui-Zhong Hu:** Project administration, Writing - review & editing. **Xian-Wu Bi:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was financially supported by CAS Hundred Talents Program to XWH, National Natural Science Foundation of China (92462306, 42473080, U23A2027, 92162323), Provincial Science and Technology Plan Project of Guizhou Province (Qiankehe Foundation-ZK[2023]Key 050), CAS “Light of West China” Program to YMM, and Special fund of the State Key Laboratory of Ore Deposit Geochemistry (202101, 202301). We thank Zhi-hui Dai and Dan Chen from the IGCAS for LA-ICP-MS analyses, and Liang Qi from the IGCAS for solution ICP-MS and ICP-OES analyses. Editors Mei-Fu Zhou, Xin-Fu Zhao and two anonymous reviewers are thanked for constructive comments.

Appendix A. Supplementary material

Table S1 Mineral compositions of different types of samples from the Makeng deposit.

Table S2 Major and trace elements in the whole rocks from the Makeng Fe deposit.

Table S3 LA-ICP-MS result (in ppm) of trace elements in the minerals from the Makeng Fe deposit.

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Figure captions

Fig. 1. (a) Simplified geological map showing the location of South China plate (modified from Shu et al., 2021; Zuo et al., 2015); (b) Geological map of the Yongmei depression zone, showing the major structures, strata and mineral deposits (modified from Zhang et al., 2015b).

Fig. 2. (a) Geological map of the Makeng Fe polymetallic deposit (modified from Zhang and Zhang, 2014); (b) Geological profile along the exploration line 61 in the Makeng deposit (modified from Zhang et al., 2012b).

Fig. 3. Field photos and photographs of hand specimens of the different types of ores from the Makeng Fe deposit. (a) The clear contact boundary between Fe ore body and mineralized skarn; (b-c) Metallic sulfide veins in Fe ore bodies; (d) Massive Fe ore composed of magnetite and andradite; (e) Disseminated Fe skarn ore with an assemblage of magnetite + quartz + amphibole + quartz; (f) Banded Fe ore, with alternated magnetite and alteration minerals such as chlorite kaolinite, and amphibole; (g-h) Banded ore, with alternated magnetite and assemblages of pyrite-pyrrhotite-chalcopryrite; (i) Disseminated Fe-Mo skarn ore containing molybdenite veins crosscutting pyroxene and calcite. Abbreviations: Mag = magnetite, Py = pyrite, Po = pyrrhotite, Ccp = chalcopryrite, Mol = molybdenite, Hem = hematite, Adr = andradite, Px = pyroxene, Amp = amphibole, Qz = quartz, Cal = calcite, Chl = chlorite, Fl = fluorite.

Fig. 4. Photomicrographs of ores under reflected/transmitted light (b-c, f, h-k) and scanning electron microscope (a, d-e, g, l). (a) Fine-grained sphalerite present in quartz veins associated with Mag-I; (b) Andradite replaced by the first generation of magnetite (Mag-I) along the margin; (c) Bands of Mag-I associated with andradite, pyroxene, and quartz; (d) The second generation of magnetite (Mag-II) showing oscillatory zoning; (e-f) Andradite replaced by Mag-II, which was further replaced by chlorite; (g) The third generation of magnetite (Mag-III) associated with pyrite and sphalerite; (h-i) Mag-III replaced or filled by pyrite and calcite; (j-l) The fourth generation of magnetite is associated with pyrite, pyrrhotite and minor fluorite and quartz. Abbreviations is shown in Fig. 3.

Fig. 5. Correlation analysis of elements in whole rocks of the Makeng Fe deposit.

Fig. 6. (a) Box plot for LA-ICP-MS trace element data showing the content of Ga and Ge in different minerals from the Makeng deposit; (b) Box plot of Ga and Ge content in different generations of magnetite from the Makeng deposit. Mineral abbreviations are the same as Fig. 3.

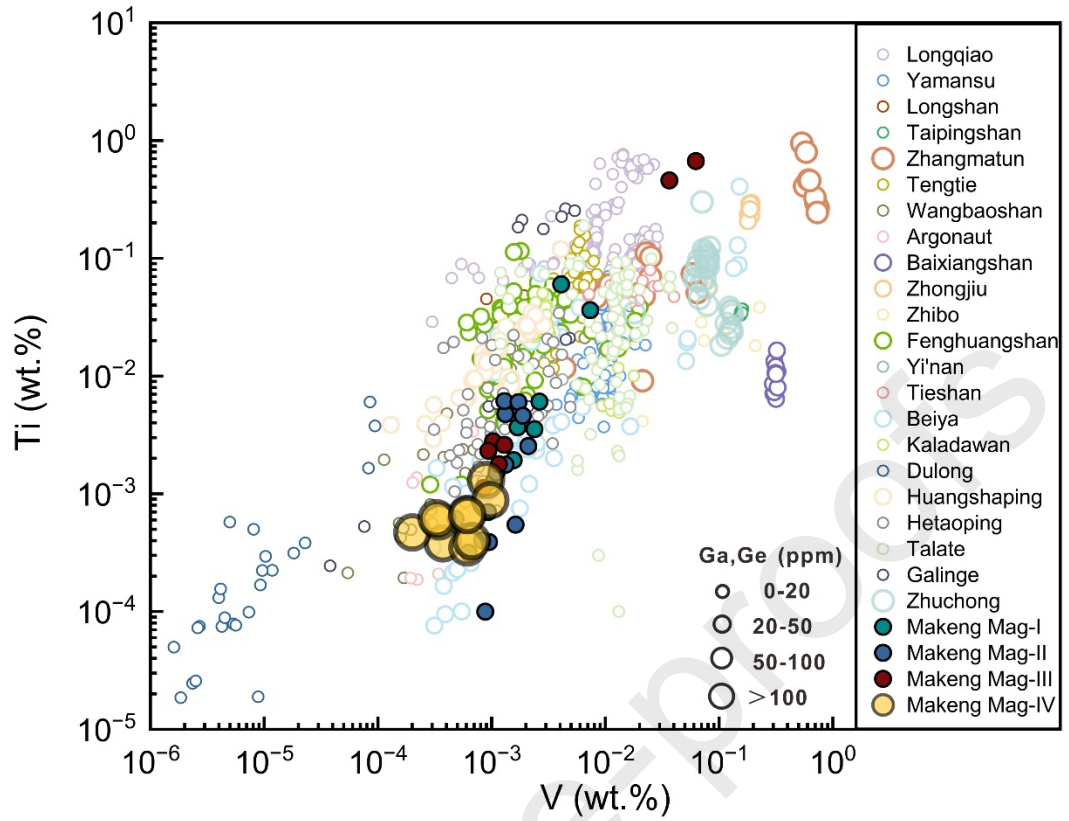
Fig. 7. Time-resolved LA-ICP-MS signal spectra for andradite, pyroxene, amphibole and magnetite in the Makeng deposit.

Fig. 8. Correlation diagrams of elements in different minerals. (a) Ga + Mn vs. Fe in andradite; (b) Ge vs. Si in andradite; (c) Ga vs. Fe in pyroxene; (d) Ge vs. Fe in pyroxene; (e) Ge + Fe vs. Al in pyroxene; (f) Ga vs. Fe in amphibole; (g) Ga + Fe vs. Mn in amphibole; and (h) Ge vs. Si in amphibole.

Fig. 9. Element correlation plots for different types of magnetite in the Makeng deposit.

Fig. 10. Comparison of Ga and Ge content in magnetite from typical hydrothermal Fe deposit, data from this study and other sources (Cai et al., 2023; Canil et al., 2016; Chen et al., 2020; Ding et al., 2018; Günther et al., 2017; Hu et al., 2017; Huang et al., 2016; Huang et al., 2018; Li et al., 2018; Li, 2012; Liang et al., 2023a; Liu et al., 2019; Sun et al., 2019; Sun et al., 2017; Wang et al., 2017a; Wang et al., 2017b; Xie et al., 2017; Zhang et al., 2023; Zhao et al., 2022; Zhao and Zhou, 2015)

Fig. 11. Binary plot of V vs. Ti in magnetite by comparing the Makeng Fe deposit with hydrothermal Fe deposits, data source is similar to that in Fig. 10.



Highlights

- Gallium enters pyroxene and amphibole by substituting $\text{Fe}^{2+}/\text{Fe}^{3+}$ or Mn^{2+} .
- Germanium enters andradite and amphibole by replacing Si^{4+} or Al^{3+} .
- Gallium in magnetite via $\text{Ga}^{3+} + \text{Al}^{3+} \leftrightarrow 2\text{Fe}^{3+}$, Ge via $\text{Ge}^{2+} \leftrightarrow \text{Fe}^{2+}$ or $\text{Ge}^{2+} + \text{Mg}^{2+} \leftrightarrow 2\text{Fe}^{2+}$.
- Content of Ga and Ge in silicates depend on crystal structure and partition coefficient.
- Enrichment of Ga and Ge in Mag-IV is due to superposition of Ga- and Ge-rich fluids.

Fig. 1

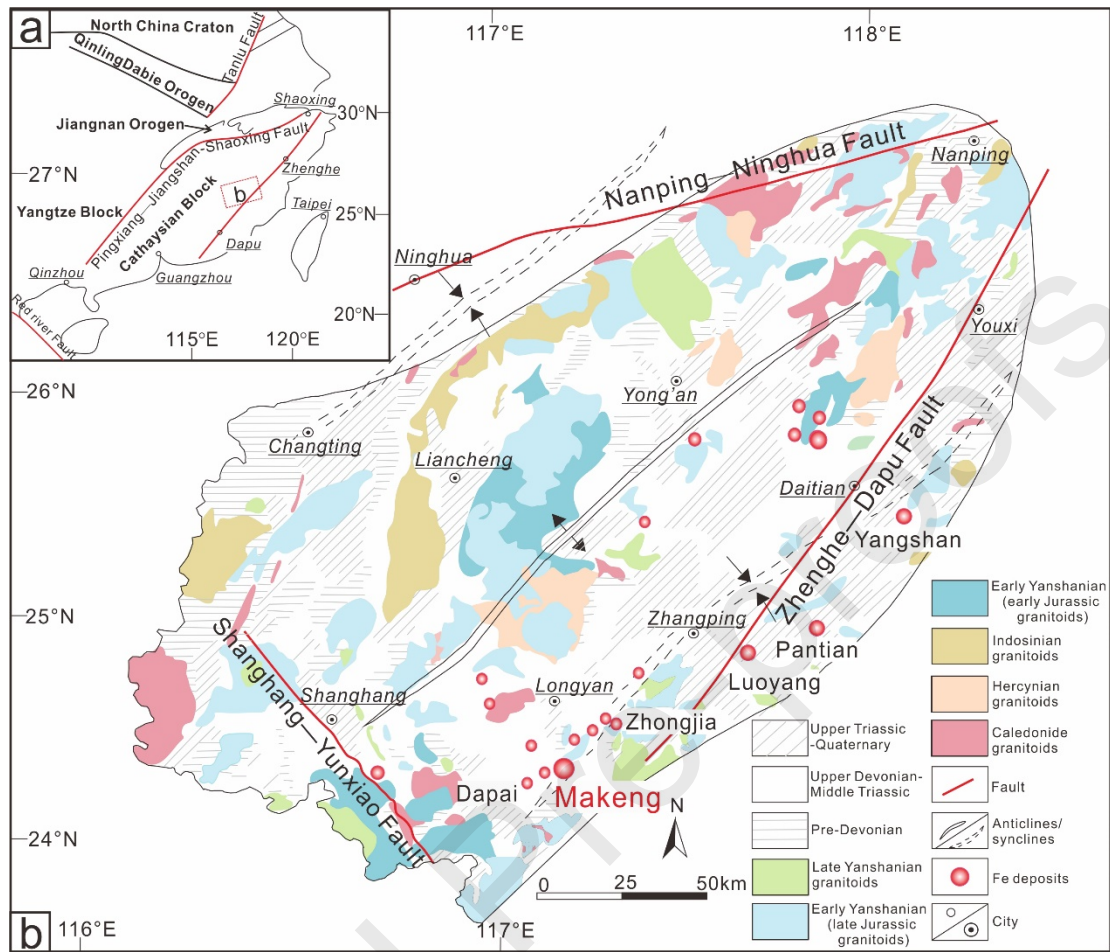


Fig. 2

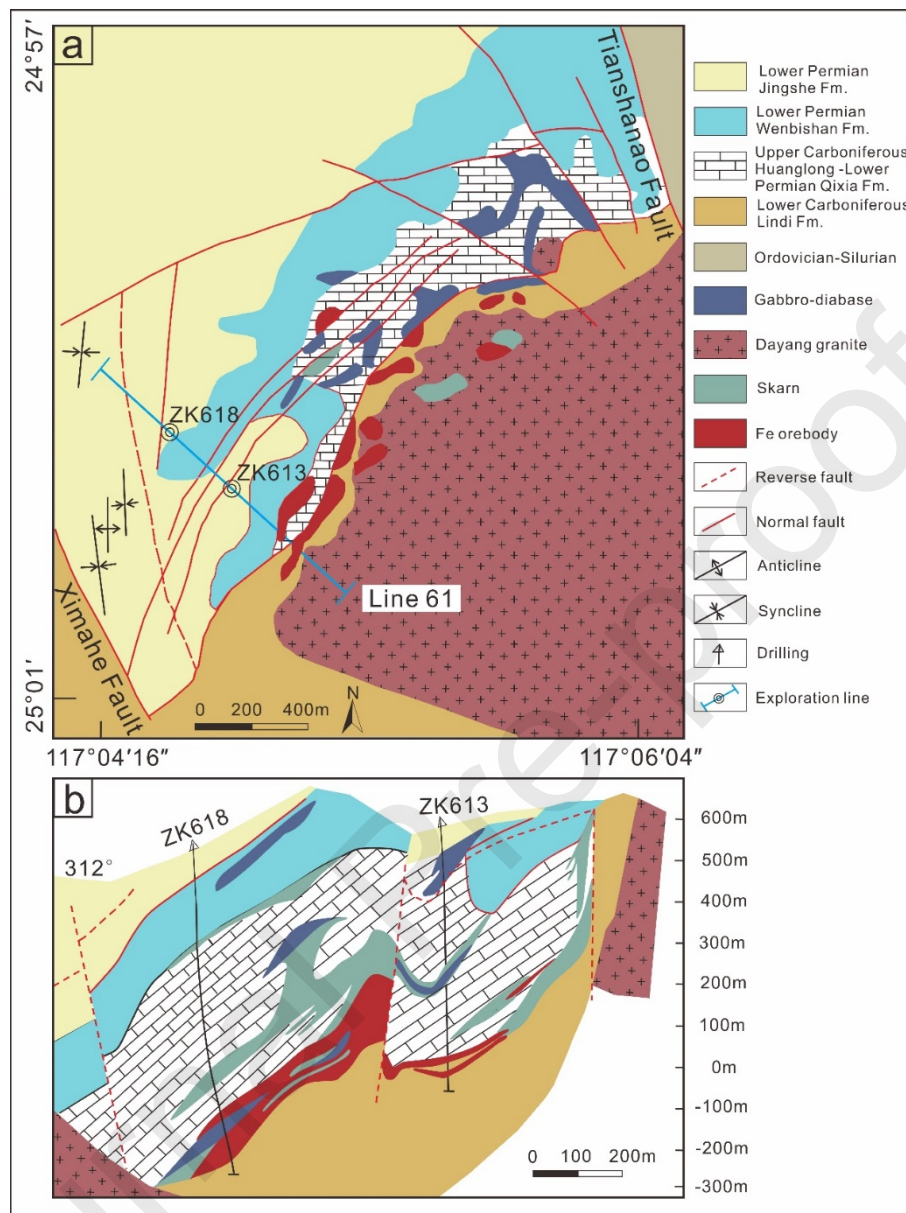


Fig. 3

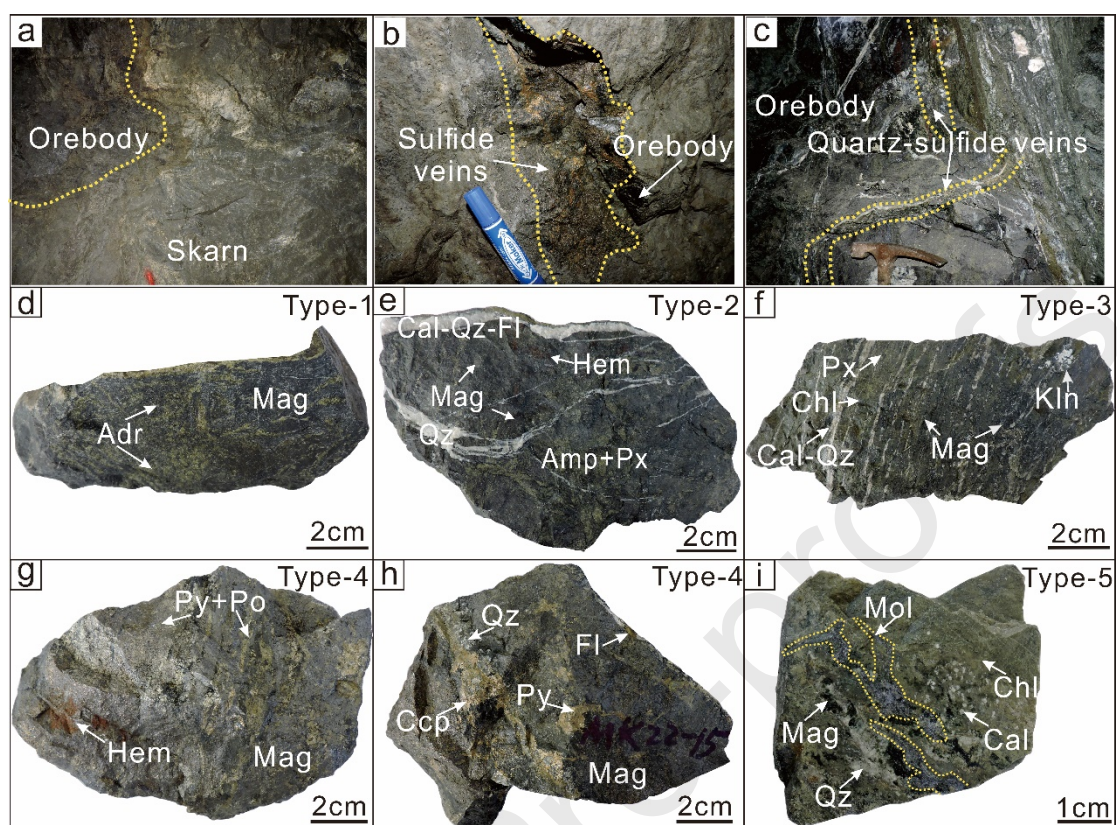


Fig. 4

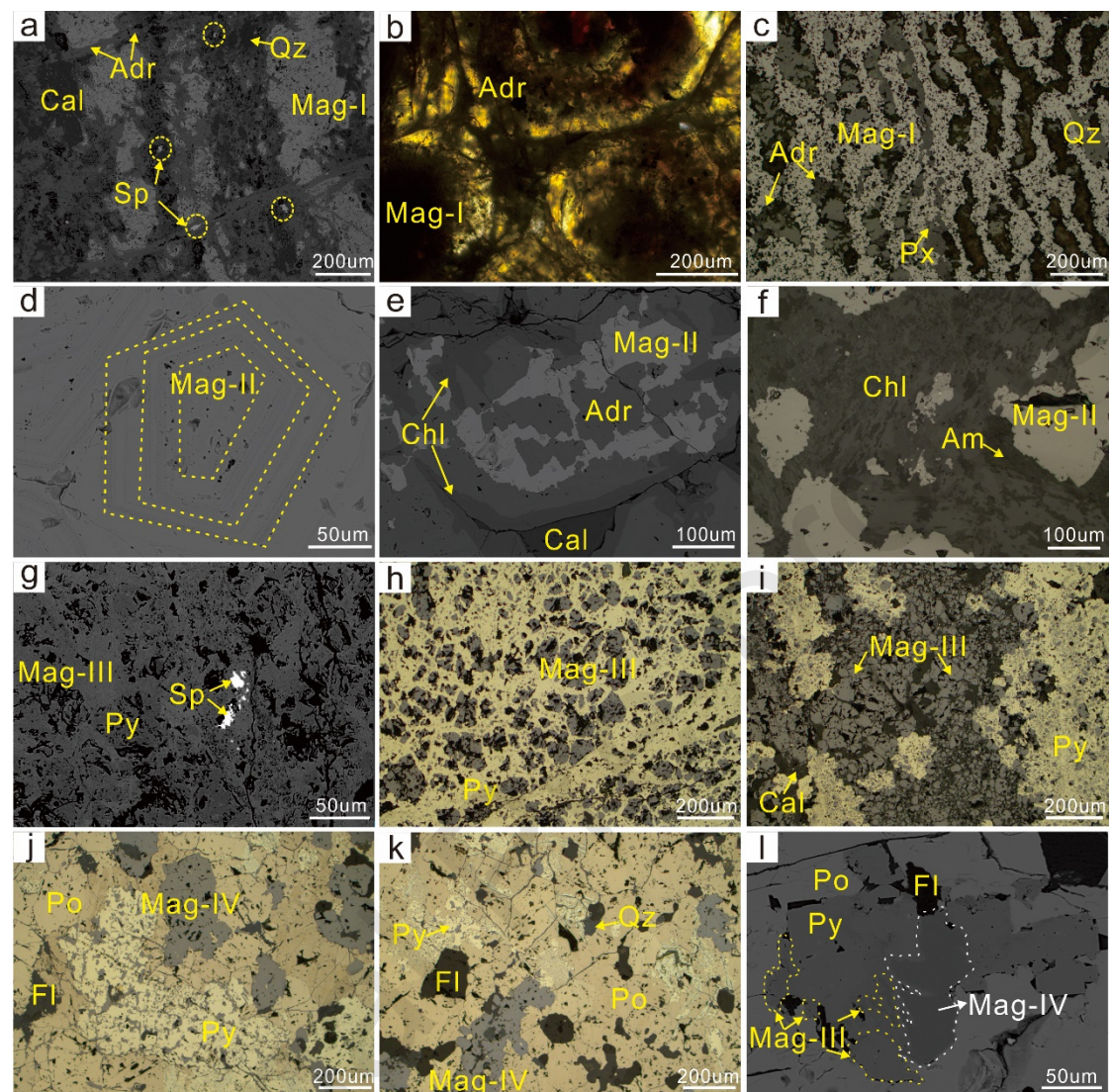


Fig. 5

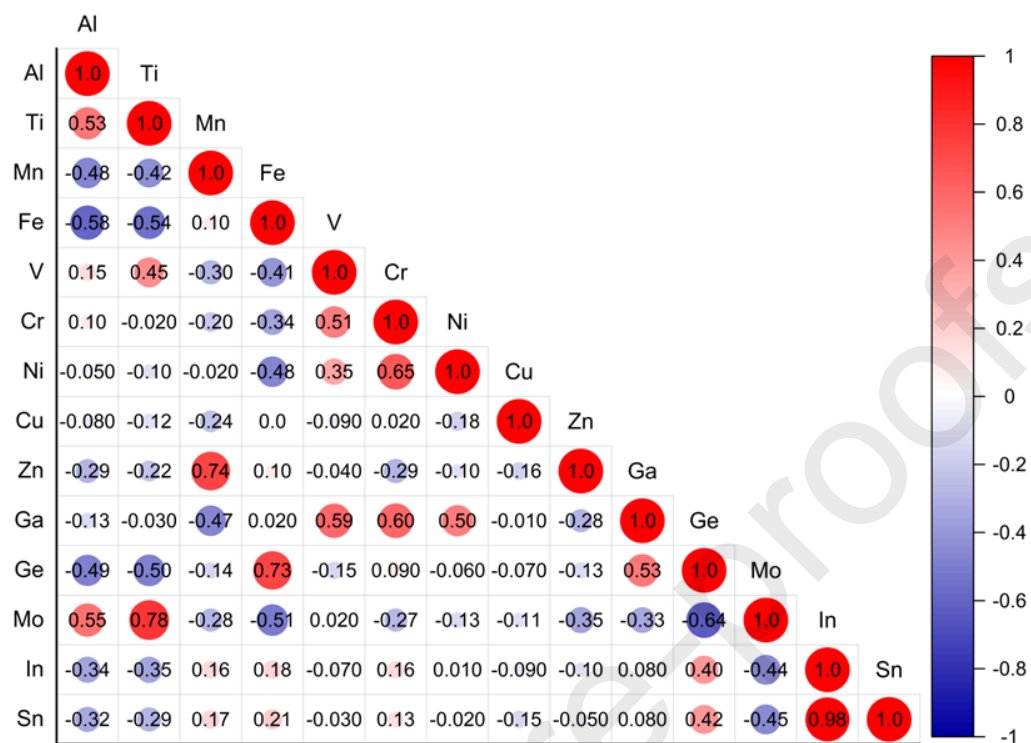


Fig. 6

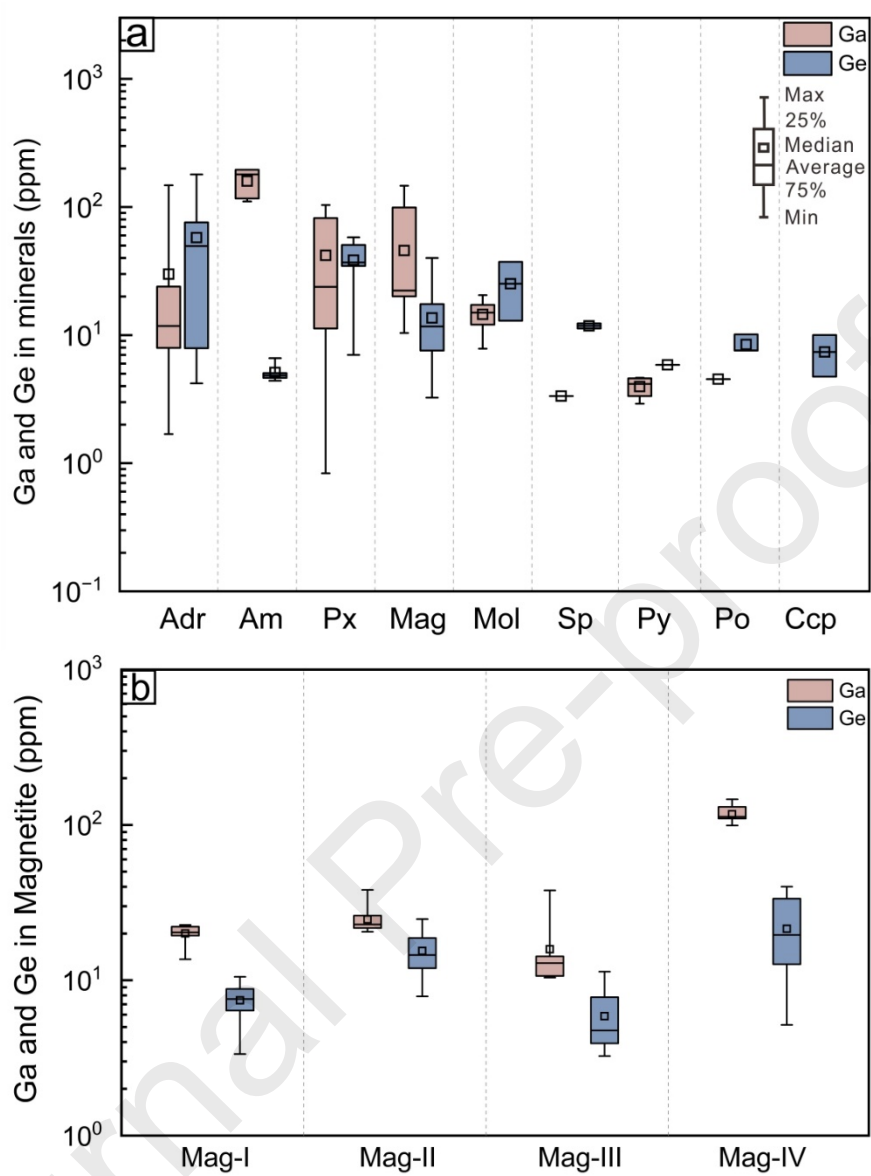


Fig. 7

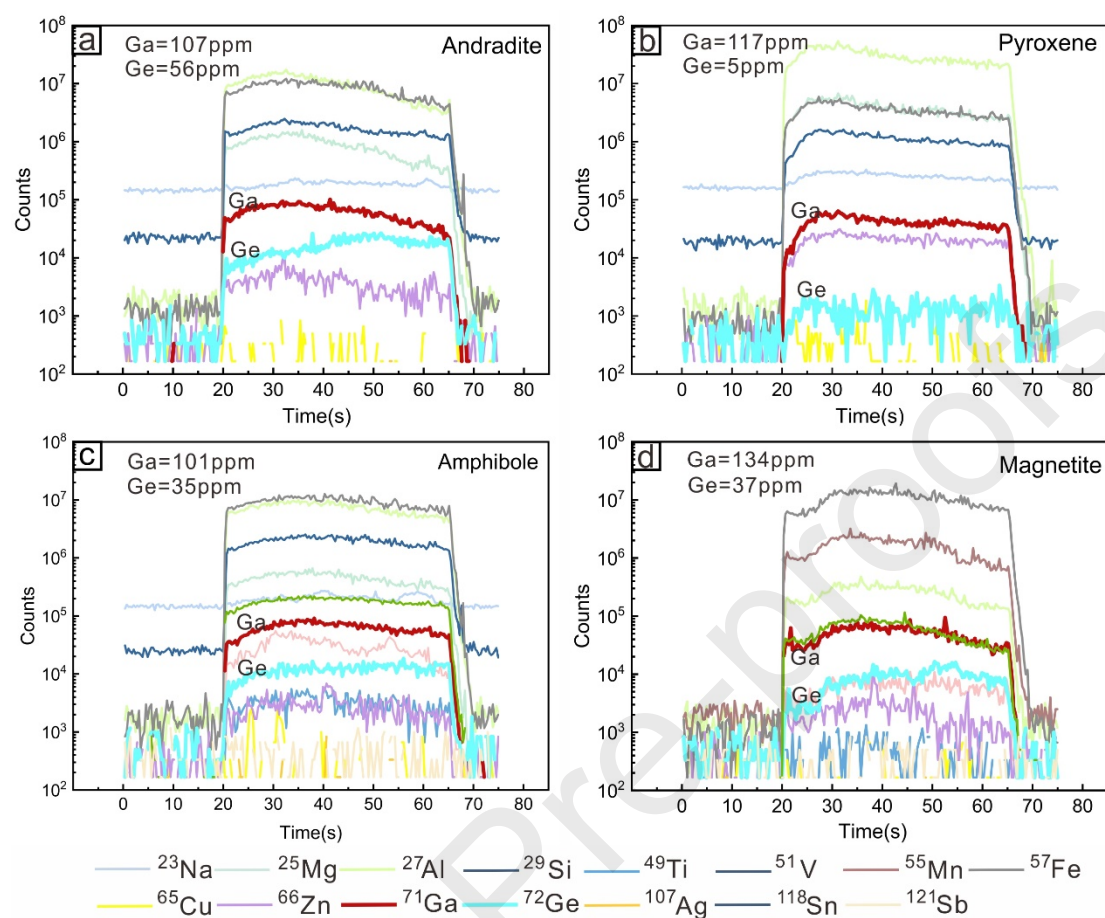


Fig. 8

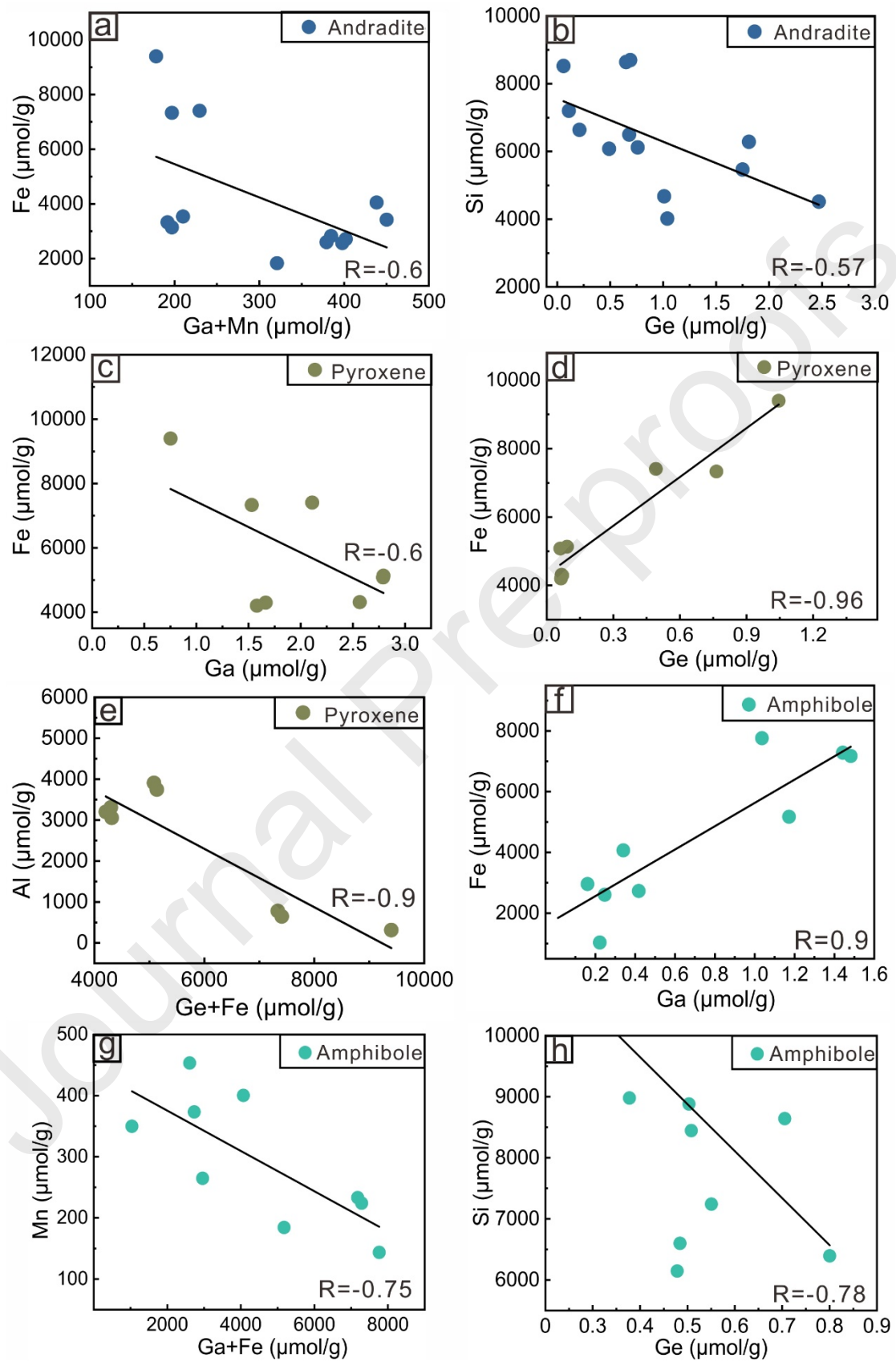


Fig. 9

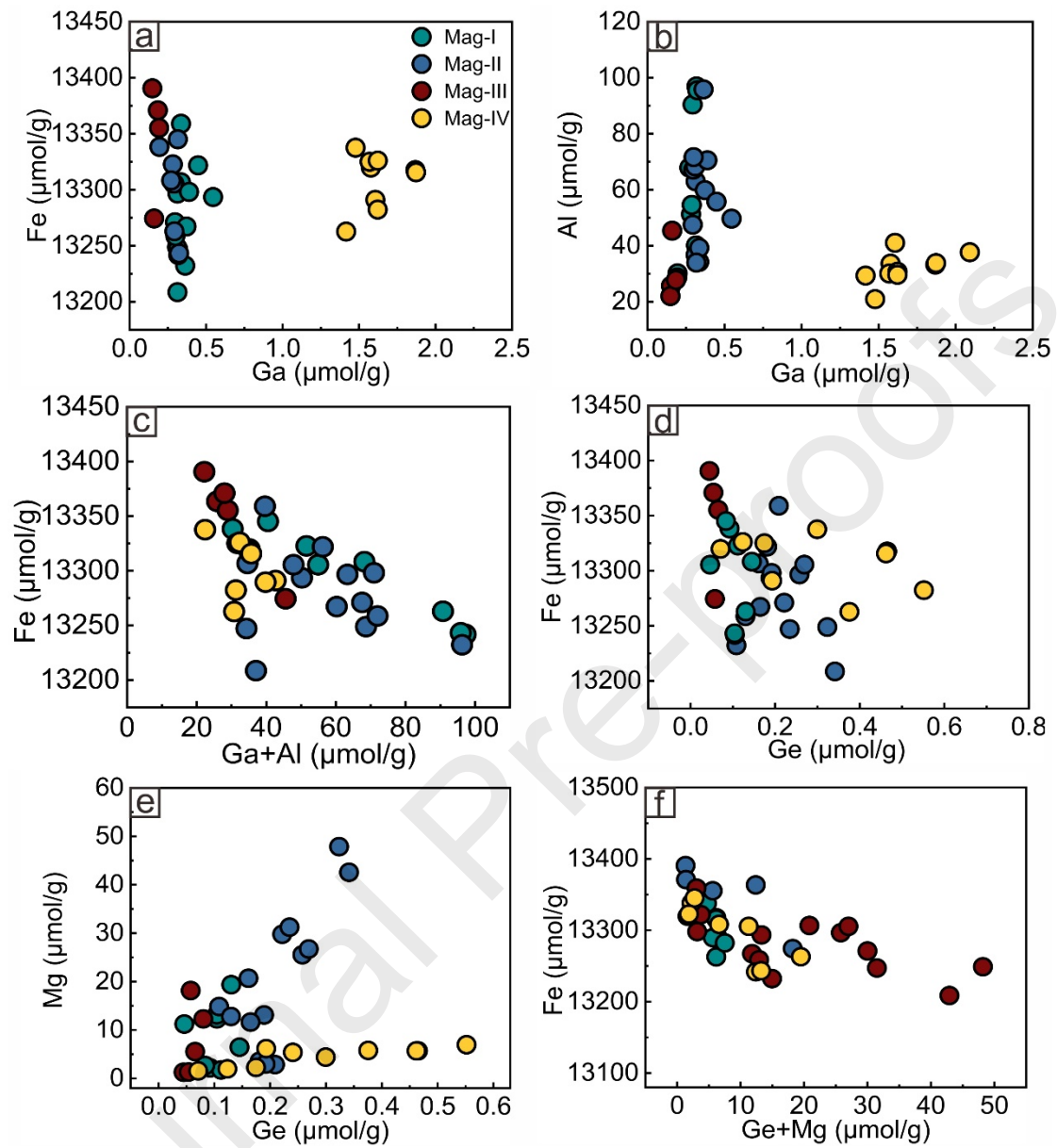


Fig. 10

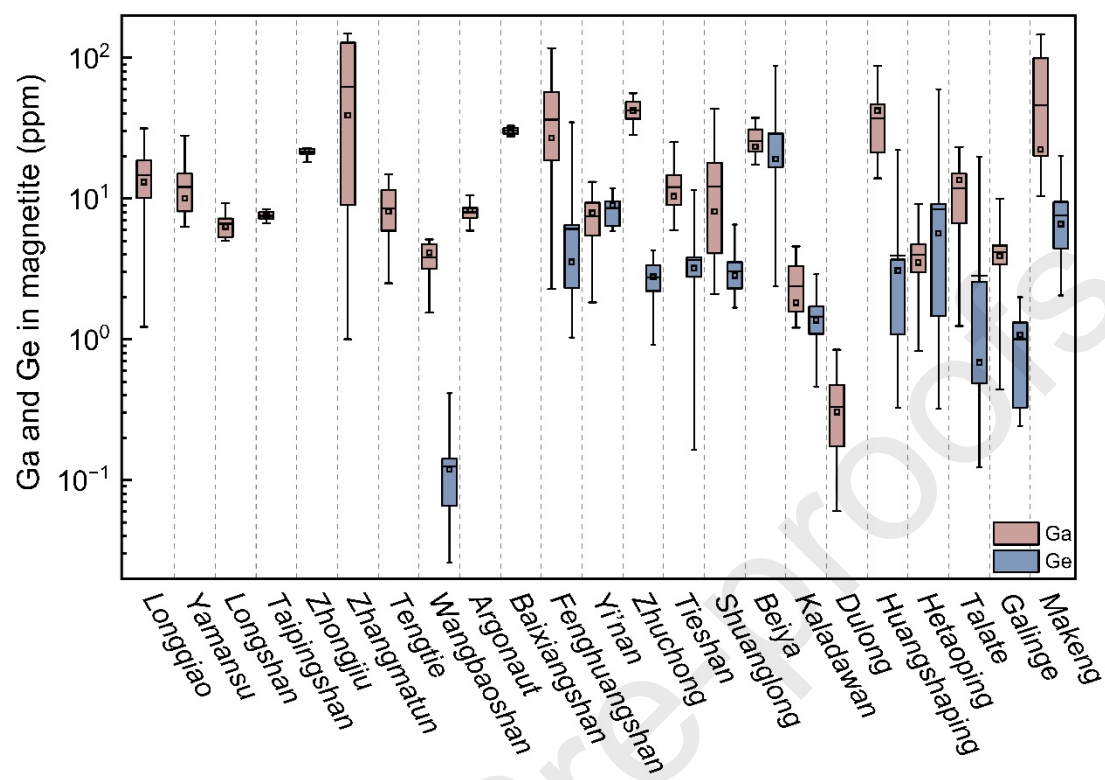
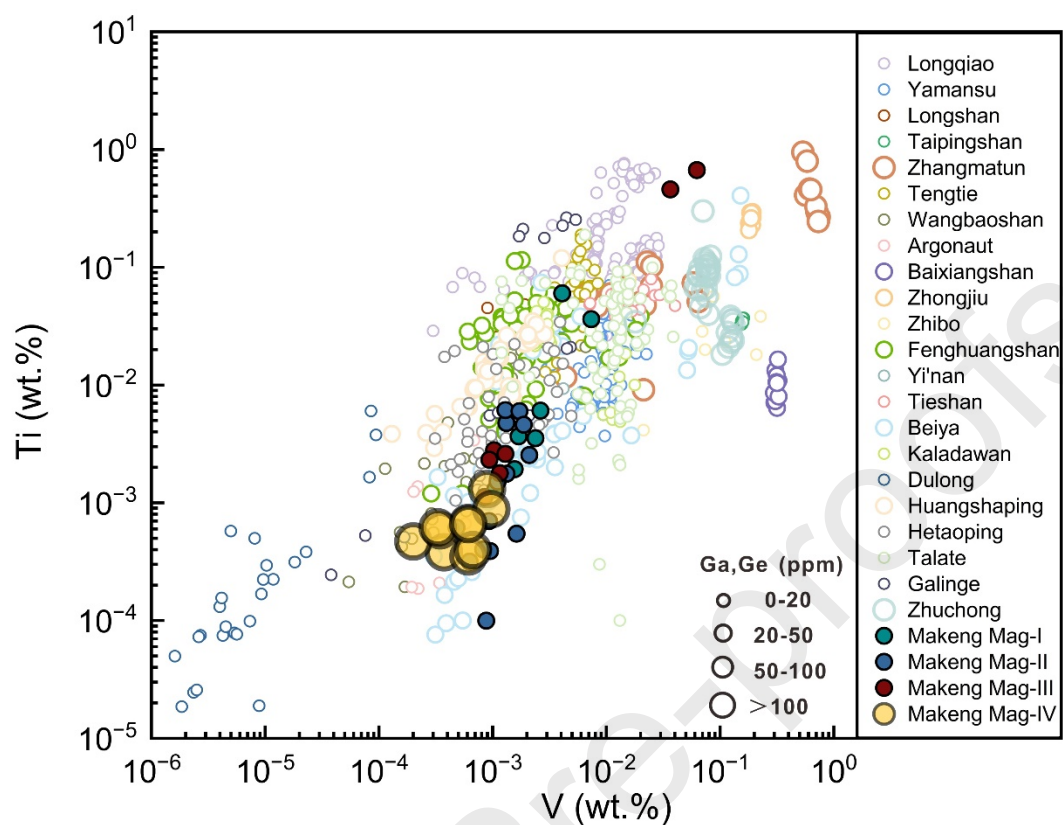


Fig. 11



Declaration of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.